

16.8 Stokes' Theorem

Prerequisite information for this section:

1. Let $\vec{F}(x, y, z) = \langle xyz, x^2yz, e^{xyz} \rangle$. Calculate $\nabla \times \vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & x^2yz & e^{xyz} \end{vmatrix} = (xze^{xyz} - x^2y)\hat{i} - (yz e^{xyz} - xz)\hat{j} + (2xyz - x^2)\hat{k}$$

2. Let $\vec{G}(x, y, z) = \langle xyz, x^2yz, 0 \rangle$. Calculate $\nabla \times \vec{G}$.

$$\nabla \times \vec{G} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & x^2yz & 0 \end{vmatrix} = (-x^2y)\hat{i} - (-xy)\hat{j} + (2xyz - x^2)\hat{k}$$

There are no new definitions in this section.

Topics to learn in this section:

1. The statement of Stokes' Theorem.
2. The connection between the curl and the circulation of a velocity field.

16.9 The Divergence (Gauss') Theorem**Prerequisite information for this section:**

1. Let $\vec{F}(x, y, z) = \langle z^2x, y^3 + \tan z, x^2z + y^2 \rangle$. Calculate $\nabla \cdot \vec{F}$.

$$\nabla \cdot \vec{F} = z^2 + 3y^2 + x^2$$

2. Let $\vec{G}(x, y, z) = \langle y^2z, x^2yz, e^{xyz} \rangle$. Calculate $\nabla \cdot \vec{G}$.

$$\nabla \cdot \vec{G} = 0 + x^2 + z + xye^{xyz}$$

There are no new definitions in this section.

Topics to learn in this section:

1. The meaning of a simple closed solid region R and its boundary surface $S = \partial R$.
2. A careful statement of the Divergence Theorem.

16.10 Summary

Prerequisite information for this section:

Give the 5 versions of the Fundamental Theorem of Calculus you have learned so far. (The first one you learned in Calculus 1, and the other four are from Chapter 16. You may want to revisit your book/ebook, sections 5.3, 16.3, 16.4, 16.8, and 16.9.)

Let $f: [a, b] \rightarrow \mathbb{R}$ be differentiable. Then

$$\int_a^b f'(x) dx = f(b) - f(a)$$

Let $C: \vec{r}(t), t \in [a, b]$ be a smooth curve. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function that is continuous over C . Then

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Let $D \subset \mathbb{R}^2$ be a closed region with $C = \partial D$ being a positively oriented, piecewise smooth simple, closed curve. If $\vec{F} = \langle P, Q \rangle$ is a vector field with continuous first order partials on D , then

$$\oint_C P dx + Q dy = \iint_D Q_x - P_y dA$$

Let S be an oriented, piecewise smooth surface with a simple, closed, piecewise smooth boundary $C = \partial S$ with positive orientation. Let $\vec{F}$ be vector field with continuous partials on an open region containing S . Then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \times \vec{F} \cdot d\vec{S}$$

Let E be a simple solid region and let $S = \partial E$ have positive orientation. Let $\vec{F}$ be vector field w/ continuous partials over an open region containing E . Then

$$\oint_S \vec{F} \cdot d\vec{S} = \iiint_E \nabla \cdot \vec{F} dV$$

There are no new definitions in this section.

Topics to learn in this section:

1. Define and demonstrate the strategy for using the methods of Chapter 16 to evaluate integrals and integral expressions.